

Geometric Flow Networks: A Physics-Informed Paradigm for Sequential Intelligence

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Abstract

Geometric Flow Networks (GFN) constitute a paradigm of neural computation that formalizes intelligence as the continuous evolution of a persistent internal world governed by structural invariants. Unlike the stateless correlation computed by self-attention mechanisms, GFN treats computation as the trajectory of a state vector flowing through a geometric manifold where inputs act as external perturbations that curve the trajectory without replacing the state. This architecture ensures that information is transformed according to structural conservation laws rather than being transiently buffered or destroyed, thereby enabling a **demonstrated** constant-memory footprint for the **Recursive State Memory** regardless of context length.

Two distinct realizations demonstrate the paradigm’s scope and generality. The Geodesic State Space Model (G-SSM) formalizes representation as a continuous flow on a learned Riemannian manifold, evolving phase-space variables $(\mathbf{x}, \mathbf{v})$ through symplectic integration. In parallel, the Inertial State Network (ISN) implements a deep stateful pipeline where both semantic scanning and world evolution maintain persistent momentum. Our empirical evaluations establish a rigorous efficiency benchmark on entry-level hardware (GTX 1650): the ISN achieves character-level language generation with a perplexity of **2.48** using only **363,329 parameters**. In long-context inference ($L = 2000$), the ISN maintains a throughput of **700 TPS**, significantly outperforming Transformer baselines which exhibit $O(N^2)$ degradation. Under optimized environments (HuggingFace Spaces), the system demonstrates a peak performance of up to **2,000 TPS**, highlighting the inherent scalability of the geodesic flow engine.

By shifting from statistical correlation to structure-preserving dynamics, GFN enables a theoretical constant state memory complexity and significant parameter reduction. While the paradigm provides **structural grounding** against hallucinations through deterministic invariants, we observe that in open-domain generative tasks, the effectiveness of this resistance is coupled to the scale and precision of the learned manifold metrics. Our findings indicate that while topological constraints offer absolute guarantees in logic, linguistic domains exhibit a “soft” resistance subject to metric resolution.

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Code and models are available at <https://github.com/DepthMuun/gfn> and <https://huggingface.co/DepthMuun>.

1 Introduction

The deep learning revolution of the past decade has been fundamentally defined by a single architectural choice: the attention mechanism. Introduced by Vaswani et al. [19], self-attention enabled unprecedented parallelization in sequence modeling. Consequently, it scaled to become the technical backbone of modern large language models, vision transformers, and multimodal architectures.

The phrase “Attention Is All You Need” proved to be a foundational insight [19]; the architecture now serves as the primary basis upon which contemporary large-scale sequence modeling rests. However, we argue that this success represents a local maximum within the statistical paradigm. The attention mechanism treats reasoning as correlation maximization throughout a sequence. We propose **Geometric Flow Networks (GFN)** as a structural alternative that shifts the computational burden from statistical sampling to geodesic integration.

As a structural paradigm, GFN shifts the computational focus from statistical correlation to state-persistent physical evolution. In this work, we present two distinct architectures that instantiate this paradigm:

1. **Geodesic State Space Model (G-SSM)**: A differential realization that formalizes the framework as a second-order manifold state space model, evolving continuous phase-space variables through symplectic integration of learned curvature.
2. **Inertial State Network (ISN)**: A continuous dynamic system that implements the same physical principles through geometric state persistence, where information is lived by a particle flowing through manifold curvature rather than stored in memory buffers.

1.1 The Statistical Foundation and Its Limitations

The attention mechanism computes a weighted sum over all positions in a sequence:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (1)$$

where queries Q , keys K , and values V are linear projections of input representations. The softmax operation converts dot-product similarities into a probability distribution, effectively treating all token relationships as potentially significant until proven otherwise. This design choice carries several fundamental consequences:

- **Quadratic Memory**: The $O(N^2)$ memory scaling of attention with respect to sequence length creates barriers that constrain practical context lengths and necessitates expensive memory management strategies like KV-caches.
- **Stateless Computation**: Each forward pass treats tokens as independent observations to be correlated rather than as states in a dynamical system. This statelessness mandates explicit memory storage through KV-caches rather than implicit state compression.
- **Probabilistic Hallucination**: Because attention operates on learned correlations without grounding in physical constraints, models frequently produce outputs that

are statistically plausible but semantically invalid. These hallucinations emerge from the system’s inability to distinguish between correlation and causality.

1.2 The Path Forward: Physics-Based Intelligence

We propose that the next paradigm shift requires abandoning the statistical foundation in favor of a physics-informed approach. Rather than merely learning token correlations, we propose learning the **structural invariants** of the domain: the fundamental laws governing valid state transitions. Within this framework, we introduce **Geometric Flow Networks (GFN)**, a paradigm in which computation is formalized as a particle flowing through a structured state space. This transition offers several theoretical advantages:

- **Recursive State Persistence:** Once the domain’s invariants are captured, information is preserved through state persistence rather than explicit history buffering. The internal world model (the latent state) operates as a persistent geometric configuration, dramatically reducing the memory overhead associated with context expansion through its intrinsic nature rather than explicit caching.
- **Architectural Agnosticism:** If the model learns the underlying physics of a domain, different signals (text, image, audio) represent distinct projections of the same invariant structure, enabling native multimodality through geometric unification.
- **Deterministic Grounding:** Physical constraints enforce validity conditions that provide a structural resistance to hallucination, moving from statistical regularizers to hard-coded geometric boundaries.

By replacing probability with structured state evolution, the **GFN Paradigm** escapes the boundaries of quadratic memory scaling and statistical hallucination. The following sections formalize this framework and demonstrate its empirical efficacy across both differential and continuous dynamic domains, establishing a deterministically stable foundation for the next generation of artificial intelligence.

2 Theoretical Foundations

2.1 The Geometric Locus of Intelligence

We begin by formalizing the distinction between statistical attention and geometric interaction. In the Transformer paradigm, an input sequence $\mathbf{x} = (x_1, \dots, x_N)$ exists in a flat embedding space where relevance is a measure of learned similarity. In this framework, reasoning reduces to computing weighted averages based on dot-product similarities.

In contrast, our framework treats inputs as forces that perturb a geometric entity flowing through a structured **geometric locus** $\mathcal{M}$ (the domain of validity). Each valid cognitive state or relationship corresponds to a point within this structured domain, and reasoning is defined as the **structural evolution** of state following geodesic trajectories through the manifold. Crucially, the GFN paradigm is agnostic to the specific mathematical formalism of $\mathcal{M}$: it may be instantiated as a Riemannian manifold with Christoffel symbols, a discrete transition graph obeying conservation laws, or a continuous flow field, provided the evolution satisfies the five defining pillars of the paradigm.

The fundamental equation governing this flow (Equation 2) is:

$$\mathbf{y} = \gamma_{\mathcal{M}}(0, 1; \mathbf{x}_0, \mathbf{v}_0) \quad (2)$$

where $\gamma_{\mathcal{M}}$ represents the geodesic trajectory on manifold $\mathcal{M}$ starting from initial position $\mathbf{x}_0$ with initial velocity $\mathbf{v}_0$, arriving at final state $\mathbf{y}$. The state does not teleport between configurations; it flows continuously through the geometry.

2.2 The Structural Evolution Principle

The GFN paradigm defines intelligence as the evolution of a state $(\mathbf{x}, \mathbf{v})$ according to a **stationary action principle**. In this view, a sequence of inputs acts as a series of perturbations (forces) that bend the geodesic trajectory of the particle through the manifold. The trajectory is governed by a generalized flow operator $\mathcal{F}$ (Equation 3):

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}, \quad \frac{d\mathbf{v}}{dt} = \mathcal{F}(\mathbf{x}, \mathbf{v}, \mathbf{f}_{ext}; \theta) \quad (3)$$

where $\mathcal{F}$ is designed to preserve specific **structural invariants** of the domain while accepting external force perturbations $\mathbf{f}_{ext}$.

In the differential realization (**G-SSM**), this evolution is formalized as a **geodesic flow** on a learned Riemannian manifold $\mathcal{M}$. The trajectory follows the geodesic equation:

$$\frac{d^2 x^i}{dt^2} + \Gamma_{jk}^i \frac{dx^j}{dt} \frac{dx^k}{dt} = 0 \quad (4)$$

where Γ_{jk}^i (Christoffel symbols) represent the learned curvature of the semantic space. Logical consistency corresponds to following the path of least resistance through continuous phase space.

Conversely, in the continuous dynamic realization (**ISN**), the evolution $\mathcal{F}$ is instantiated through **continuous geodesic flow** where the state’s momentum carries it forward even when no external forces are applied. The key distinction from G-SSM is that ISN eliminates discrete timesteps entirely, treating state evolution as a continuous differential flow governed by three principles:¹ (1) **State Inertia**: the world persists without external clock synchronization; (2) **Geometric Coupling**: inputs curve the trajectory without replacing state; (3) **Manifold Omnipresence**: all computation occurs within invariant-preserving geometry.

2.3 Conservation Laws and Noether’s Theorem

The stability of trajectories on $\mathcal{M}$ is guaranteed by embedding conservation laws directly into the architecture. In physical systems, conserved quantities (mass, energy, momentum) constrain possible evolutions. According to Noether’s Theorem [14], every differentiable symmetry of a physical system’s action corresponds to a conservation law.

By designing the manifold $\mathcal{M}$ to possess specific geometric symmetries, we ensure that the corresponding conservation laws are hard-coded into the model’s dynamics. For example, in the domain of integer parity (XOR), the parity of the world state must be conserved across operations. This stands in contrast to statistical attention, where constraints are limited to soft probabilistic normalization. In a geometric framework,

¹This represents a fundamental departure from discrete recursions ($h_{t+1} = f(h_t, x_t)$) to continuous vector fields ($\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, u(t))$), enabling temporal resolution independence where the state can be queried at arbitrary time points (e.g., $t = 1.5, t = 1.0001$); this constitutes a theoretical capability of the continuous GFN paradigm beyond discrete token-based implementations.

violations are not merely unlikely: they are geometrically inconsistent with the chosen topology, making them structurally impossible rather than statistically improbable:

$$\sum_i \psi_k(\text{input}_i) = \text{State}_k(\text{world}) \quad (5)$$

where ψ_k represents a conserved property mapping and the equality is enforced by the geometry rather than learned through statistics.

3 The Five Pillars of Geometric Flow Networks

The **Geometric Flow Networks (GFN)** paradigm is defined by five structural pillars that capture its philosophical essence rather than prescribing specific implementations. These pillars distinguish GFN from both statistical models (attention, SSMs) and generic continuous models (Neural ODEs) by articulating what the paradigm fundamentally *is*, not how it must be *computed*:

1. **Geodesic State Flow:** The state evolves as a continuous trajectory through a learned geometry, not as a statistical transformation of tokens. This pillar captures the essential nature of GFN computation: the system computes transitions as flow along geodesics in a manifold where valid transitions correspond to trajectories of minimal resistance. Unlike attention which computes weighted correlations, GFN computes how state moves through semantic curvature. The state possesses “semantic inertia”: history manifests as trajectory shape, not as explicit storage.
2. **Persistent Internal World:** The state exists as a geometric configuration that persists independently of inputs; inputs perturb the trajectory without replacing the state. This pillar articulates an ontological distinction: the internal world is not a memory buffer that stores history (like a KV-cache), but a reality that *is*. Inputs do not add information to a list; they curve the space-time where the state orbits. A Transformer without KV-cache forgets everything; the GFN world exists as geometry itself, not as a record of events.
3. **Structural Invariance:** At least one conservation law (physical, logical, or topological) governs valid transitions, making certain states structurally impossible rather than merely improbable. This is the paradigm’s most philosophically distinctive pillar. Invariants are not soft regularization or probabilistic normalization (like softmax in attention), but physical laws encoded in geometry. In logical domains (XOR), the space has toroidal topology: invalid transitions are geometrically impossible. In semantic domains, invariants are “soft” but remain structural, not statistical.
4. **Causal Locality:** Dynamics emerge from local interactions (forces, curvature, couplings), not from global correlation over the entire sequence. This pillar distinguishes GFN from architectures requiring simultaneous access to all historical context. Locality here is not necessarily spatial (as in CNNs) but causal: the next state depends on forces acting on the current state, not on computing similarity with all prior tokens. This enables memory without memory buffers.
5. **Physics-Grounded Computation:** Validity constraints are geometric and topological, not statistical; coherence is measured in terms of curvature and conservation, not likelihood. This pillar articulates that GFN is not “more of the same with a different name”. Constraints are not learned statistics or probabilistic normalization,

but validity conditions encoded in geometry. The system “knows” which states are invalid because they are topologically inconsistent, not because they are statistically improbable.

Two architectures demonstrate the paradigm’s scope: G-SSM operates in continuous phase space with explicit symplectic integration, while ISN operates as a continuous dynamic system with implicit geodesic flow. Both satisfy all five pillars by capturing their philosophical essence rather than adhering to specific algorithmic constraints.

3.1 Realization A: Geodesic State Space Model (G-SSM)

The **G-SSM** is the discrete-integration realization of the GFN paradigm. It satisfies the five pillars through the following mechanisms:

- **Adaptive Riemannian Metrics:** Dynamic estimation of the domain’s metric tensor, allowing the model to non-linearly transform its tangent space to match the logical requirements of the token sequence.
- **Stability Sidecars:** Specific implementations of G-SSM incorporate *Singularity Gating* and *Numerical Hysteresis* to manage finite-precision effects, along with *Holographic Dynamics* for associative state persistence.

The G-SSM treats each input token as an external force that perturbs the geodesic flow. The state’s trajectory is determined by the competition between its inherited momentum and the curved geometry of the manifold, with external inputs bending but not replacing the configuration.

3.2 Realization B: Inertial State Network (ISN)

The **ISN** represents a continuous dynamic realization of the GFN paradigm that satisfies all five pillars through its unique architectural design. Unlike G-SSM, which operates through discrete symplectic integration steps, the ISN eliminates fixed timesteps entirely, treating state evolution as a continuous differential flow governed by geodesic inertia. In this framework, information is not stored but lived: state flows through manifold curvature, and the past shapes the future not through accumulation but through the way history has molded the trajectory.

The ISN is architecturally defined by three core modules that implement the five GFN pillars:

1. **GFNScanner:** An embedding layer followed by flow dynamics via $\tanh(W \cdot s)$, where s is the current state. This module encodes tokens as impulses and maintains persistent semantic momentum.
2. **GFNWorld:** The core state evolution engine implementing geodesic flow through drift (inertia) and diffusion (external impulses). The update rule is:

$$\mathbf{W}_{t+1} = \mathbf{W}_t + \text{drift}(\mathbf{W}_t) + \text{diffusion}(\mathbf{f}_{ext}) \quad (6)$$

where $\mathbf{W}_t$ is the persistent world state, ensuring $O(1)$ memory complexity through state persistence rather than history accumulation.

3. **ThresholdEmitter:** The materialization layer that projects world states to token logits through learned thresholds, completing the pipeline from geometric state to discrete output.

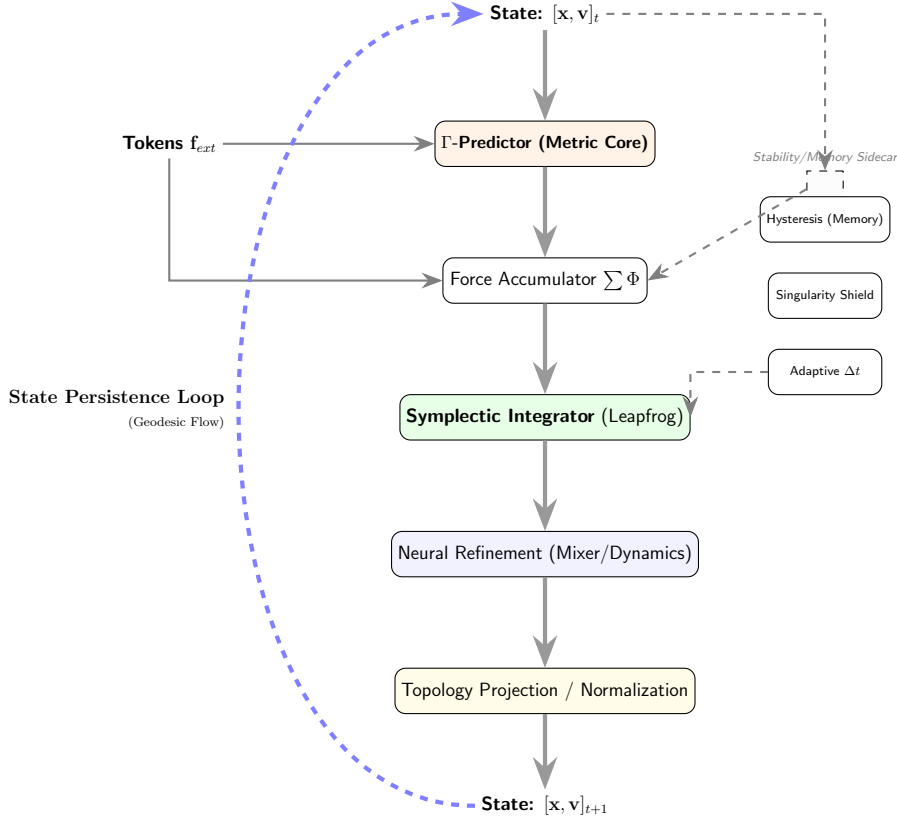


Figure 1: High-Fidelity G-SSM Architecture: The semantic state is propagated through a symplectic geometric engine, refined by a neural dynamics filter, and projected onto the manifold topology. The "Persistence Loop" ensures that token-induced forces update continuous phase-space variables (position and momentum) rather than transient embeddings. The optional **sidecar modules** (dashed box) are experimental components for stability research; the system is capable of learning the underlying domain physics natively via the core geodesic engine.

In the current autoregressive realization, the ISN flows through geometry at discrete intervals dictated by high-frequency sampling (the tokenizer), though the internal dynamics admit arbitrary temporal resolution. The detailed internal architecture is illustrated in Figure 2.

By prioritizing continuous geodesic flow over discrete token processing, the **ISN** achieves what we term **Inertia-Grounded Intelligence**: a form of computation where reasoning emerges from the geometry of a persistent internal world rather than from statistical associations between discrete tokens.

4 The Computational Advantage

4.1 Complexity Analysis: $O(1)$ vs. $O(N^2)$

The computational complexity distinction between attention and geometric interaction has profound practical implications for memory scaling and inference efficiency.

Notes: The **ISN** forward pass is $O(N)$ in terms of total sequential latency, reflecting the inherent causality of generation. However, the **per-token step complexity** is strictly $O(1)$ with respect to context length, as the model avoids the quadratic re-processing of history through its persistent state. We distinguish between the **Recursive State**

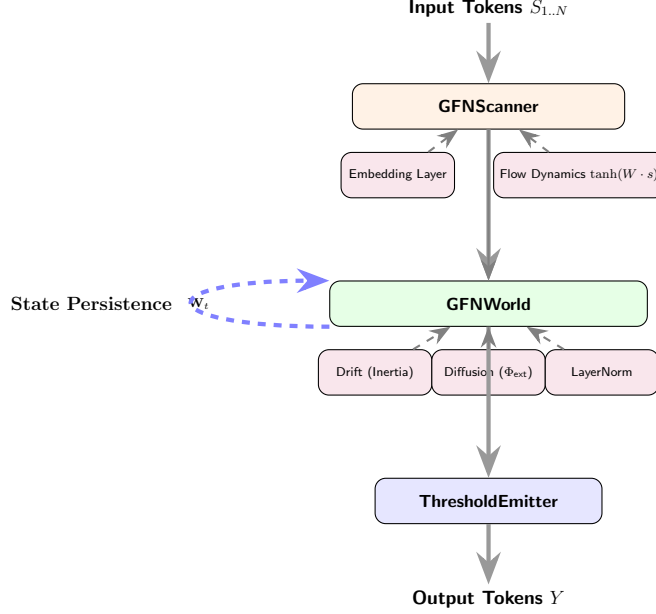


Figure 2: ISN Architecture: The ISN implements three core components: (1) **GFNScanner** embeds tokens and applies flow dynamics via $\tanh(W \cdot s)$; (2) **GFNWorld** evolves the persistent state $\mathbf{W}$ through drift (inertia) and diffusion (external impulses) with $\mathbf{W}_{t+1} = \mathbf{W}_t + \text{drift}(\mathbf{W}_t) + \text{diffusion}(\mathbf{f}_{ext})$; (3) **ThresholdEmitter** materializes world states to logits. The state persistence loop ensures $O(1)$ memory complexity. Unlike G-SSM which uses symplectic integration, ISN employs continuous differential flow where momentum carries state forward without discrete timesteps at the dynamical level.

Memory ($O(1)$), which represents the core persistent world model, and the **Processing Buffer** ($O(N)$) used in current batch implementations for gradient accumulation and monitoring. The key insight is that once the structural invariants are learned, updating the trajectory requires only the current state configuration. We distinguish between **State Memory Persistence** ($O(1)$), and the **Emission Buffer** ($O(N)$), which gathers generated tokens for external decoding.

Table 1: Complexity comparison between attention mechanism and geometric interaction.

Metric	Attention	ISN (GFN)
Batch Forward (N tokens)	$O(N^2 \cdot d)$	$O(N \cdot d)$
Step Complexity (per token)	$O(N \cdot d)$	$O(1)$
Inference Memory	$O(N \cdot d)$	$O(1)$
Inference Speed ($L = 2000$, GTX 1650)	231 TPS	700 TPS
Training Memory	$O(N \cdot d)$	$O(1)^\dagger$

[†]The GFN paradigm achieves $O(1)$ through the Adjoint Sensitivity Method. Current modular implementations exhibit $O(N)$ scaling due to gradient accumulation in standard backpropagation, though with several orders of magnitude lower constants than attention.

Furthermore, G-SSM avoids the $O(d^3)$ complexity of brute-force Christoffel computation by utilizing a **Low-Rank Riemannian Decomposition**:

$$\Gamma_{ij}^k \approx \sum_{r=1}^R W_{rk}(U_{ir}U_{jr}) \quad (7)$$

where $R \ll d$. This reduces the contraction cost to $O(d^2R)$, making the system scalable to high dimensions ($d > 512$) comparable to Transformer blocks, while naturally regularizing the metric manifold through the low-rank constraint to ensure positive-definite stability.

Furthermore, unlike standard backpropagation through time (BPTT), which requires storing all intermediate activations ($O(N)$ memory), geometric models can utilize the **Adjoint Sensitivity Method** [3, 15]. Formally, the method computes the gradient of a loss $\mathcal{L}$ with respect to the state $\mathbf{x}(t)$ by solving an adjoint differential equation backwards in time:

$$\frac{d\mathbf{a}}{dt} = -\mathbf{a}(t)^T \frac{\partial \mathcal{F}}{\partial \mathbf{x}} \quad (8)$$

where $\mathbf{a}(t) = \partial \mathcal{L} / \partial \mathbf{x}(t)$ is the adjoint state. This allows computing gradients by solving a second, adjoint ODE backwards in time, reducing training memory complexity to $O(1)$ with respect to sequence length. This makes **G-SSM** uniquely suitable for training on massive contexts (e.g., 1,000,000+ tokens) that would be physically impossible for attention-based architectures to fit in VRAM. Regarding sequential latency, GFN accepts $O(N)$ temporal processing as a fundamental tradeoff for long-horizon persistence, prioritizing structural integrity over the burst-parallelization of stateless models.

4.2 Architectural Modality-Agnosticism Through Geometric Unification

A notable advantage of the geometric approach is **modality-agnosticism**: the ability to support heterogeneous inputs without core architectural modification. Different modalities appear as distinct projections of the same underlying structural reality.

An image of a cat and the text “cat” both correspond to the same concept in the semantic manifold. Statistical models must learn this correspondence through joint training on paired data: a soft association that can fail when faced with novel combinations. Geometric models can represent the concept as a single point on the manifold, with different modalities providing different coordinate representations of the same geometric entity:

$$\text{Cat}_{\text{image}} = \text{Proj}_{\text{image}}(\mathbf{p}) \quad (9)$$

$$\text{Cat}_{\text{text}} = \text{Proj}_{\text{text}}(\mathbf{p}) \quad (10)$$

$$\text{Cat}_{\text{audio}} = \text{Proj}_{\text{audio}}(\mathbf{p}) \quad (11)$$

where $\mathbf{p}$ is the invariant geometric representation and Proj represents the modality-specific projection. The GFN paradigm proposes that what is shared across modalities is not merely a static point in space, but the **dynamical invariants** (the physics engine) that govern how those points evolve. Once the underlying transition laws are established, modality transitions become equivariant transformations rather than complex mapping problems.

4.3 Out-of-Distribution Generalization

Statistical models generalize through interpolation in high-dimensional embedding space. This generalization is inherently limited by the training distribution: any input outside this distribution requires extrapolation, which statistical models perform poorly.

Geometric models generalize through extrapolation along learned manifolds. The key distinction is philosophical rather than merely technical. A statistical model asks “what have I seen before?” A geometric model asks “what is physically possible given the invariants that govern this domain?” When asked about symbolic logic or sequence patterns not seen in training:

- A statistical model outputs the most probable continuation based on observed patterns, which may fail when patterns deviate from training distribution.
- A geometric model computes the unique result consistent with the conservation laws defining the manifold, which must hold regardless of training distribution.

This distinction is fundamental. Statistical models can only reproduce patterns; geometric models can compute novel combinations because they understand the underlying structure governing valid states. The particle flows along geodesics determined by geometry, and these geodesics exist independently of what combinations were observed during training.

5 The Free Energy Principle and Computational Neuroscience

5.1 Connection to Friston’s Free Energy Principle

Our framework resonates strongly with Karl Friston’s Free Energy Principle [7], which proposes that all biological systems minimize free energy to maintain homeostasis. Under this framework, perception is inference about hidden states of the world, and action serves to minimize surprise (the difference between predicted and observed sensory inputs).

We argue that geometric interaction directly implements free energy minimization:

1. **Prediction:** The geometric manifold defines possible states; predictions correspond to the most likely point on this manifold given current momentum and position.
2. **Inference:** Observing new data constrains the feasible region on the manifold through force perturbations; inference narrows to states consistent with both prior geometry and new observations.
3. **Action:** Interventions test predictions; the system updates its geodesic trajectory to minimize discrepancy between predicted and observed states.

The attention mechanism, by contrast, treats prediction as pattern matching: a fundamentally different computational strategy that lacks the geometric grounding necessary for structural inference. Geometric models implement active inference through continuous flow; statistical models implement passive pattern completion through weighted averaging.

5.2 Implications for Neuromorphic Computing

The computational structure of geometric interaction aligns naturally with physical computing substrates. This alignment suggests future hardware directions that we believe will prove more efficient for geometric models:

- Attention requires random access to arbitrary memory locations (the KV-cache) and dense matrix multiplication: operations well-suited to digital computers but poorly suited to analog or neuromorphic systems.

- Geometric interaction requires integration along flow fields: operations that we believe can be naturally implemented as analog differentiation on neuromorphic substrates.

We believe that the shift to geometric intelligence may be accompanied by a shift in hardware architecture. Neuromorphic chips (Intel Loihi, IBM TrueNorth) that naturally support continuous-time dynamics could more efficiently implement geometric models than the digital architectures optimized for attention. The state vector flowing through manifold curvature maps naturally onto physical dynamics in analog hardware.

5.3 The Nature of Semantic Invariants

A fundamental challenge to the GFN paradigm is the definition of physical invariants in non-physical domains like human language. We distinguish between **Hard Topological Invariants** and **Soft Ontological Invariants**.

In the **Logic (XOR)** domain, the invariant is *Toroidal Adjacency*: a hard topological constraint where the geometry represents the modular parity of states. In this regime, violations are mathematically impossible as the manifold topology does not admit non-parity-preserving trajectories.

In the **Language (ISN)** domain, the invariant is **Ontological Coherence** (Ω). Unlike the absolute constraints of XOR, Ω is enforced through a multi-dimensional penalty function $\mathcal{V}$ that keeps entity properties $\mathbf{p} \in \mathcal{P}$ stable under the continuous vector field $\mathcal{F}$:

$$\Omega(\mathbf{W}) = \int_0^T \|\nabla_{\mathbf{W}} \mathcal{V}(\mathbf{W})\| dt \approx 0 \quad (12)$$

where $\mathcal{V}$ penalizes transitions into ontologically inconsistent geometry (e.g., a "Person" entity losing "speaking" capability). While GFN realizations provide a structural bias towards consistency, the "hardness" of this resistance in open-domain generative tasks is a function of the learned curvature precision rather than a topological guarantee.

While Neural ODEs often rely on high-order adaptive solvers (e.g., RK45), the GFN realizations prioritize **Symplectic Fixed-Step Integration**. In the **G-SSM** realization, this is augmented by **Singularity Gating**. Near singularities where the metric determinant $\det(g) \rightarrow 0$, Christoffel symbols can diverge. G-SSM implements a smooth damping mechanism (the *Singularity Gate*) that uses a sigmoid potential to reduce velocity and force in these regions, preventing numerical explosions (NaN). This stability is further reinforced by **Numerical Hysteresis** (0.05 – 2.0), which acts as a thermal stabilizer to absorb floating-point errors in sequence lengths exceeding 32,000 tokens. Such corrections acknowledge the realities of finite-precision physical simulation within the Riemannian framework.

6 Related Work and Differentiation

6.1 Linear Attention and State Space Models

Recent work has explored linear attention variants [13], state space models [9, 10], and other recurrent alternatives such as RWKV [11] and RetNet [12]. These approaches achieve $O(N)$ or $O(1)$ complexity but retain the statistical foundation: they still compute weighted sums over past positions, albeit more efficiently.

Our approach differs fundamentally. Rather than optimizing statistical attention or recurrence, we replace statistical computation with geometric integration. Linear models compute statistical associations based on history; we evaluate configurational validity based on conservation laws. This distinction concerns the fundamental nature of computation rather than marginal efficiency gains.

Furthermore, while Mamba and similar selective state space models achieve linear complexity, they maintain discrete state transitions where each step replaces or updates state based on input. The ISN differs by eliminating discrete transitions entirely, treating state evolution as continuous geodesic flow where momentum carries the particle forward and inputs merely perturb the trajectory.

6.2 Neural ODEs and Continuous Models

Neural ordinary differential equations [3] and their extensions [6] provide the mathematical framework for continuous-time neural networks. Our work extends this framework by proposing that the ODE dynamics should encode physical laws rather than arbitrary learned transformations.

However, we distinguish GFN from generic Neural ODEs through four key structural differences that ensure the flow is geodesic rather than merely continuous:

- **Geodesic vs. Generic Vector Fields:** While a Neural ODE learning a generic $f(\mathbf{x}, \theta)$ can drift into semantically invalid regions, GFN restricts the vector field to the geodesics of the manifold defined by learned Christoffel symbols Γ . The state does not move "by chance" but follows the path of least resistance through learned semantic geometry.
- **Conservation vs. Dissipation:** Standard Neural ODEs with generic integrators (Euler, RK4) can dissipate information or suffer from exploding/vanishing dynamics. GFN uses **Symplectic Integration** (Hamiltonian flow) to preserve volume in phase space, ensuring that information is conserved indefinitely during the trajectory.
- **Second-Order Momentum:** Most Neural ODEs are first-order systems ($\dot{\mathbf{x}} = f$). GFN is inherently second-order ($\ddot{\mathbf{x}} = f$), capturing inertial momentum. In GFN, the state has "semantic velocity"; history is carried forward by inertia even when no external forces are applied, whereas first-order systems lack this persistent momentum.
- **World Coherence (ISN):** ISN extends the continuous flow paradigm into a physics engine for a world-state with entity properties. The flow updates a configuration of properties that must satisfy logical coherence, moving from mere vector transformation to structured simulative reality.

Neural ODEs have been applied to sequence modeling [16] as an alternative to attention. However, these applications typically treat the ODE as a drop-in replacement for attention while retaining the statistical paradigm. We propose that the key advantage of continuous models lies not in computational efficiency but in their ability to naturally encode physical constraints that enable true generalization beyond training distribution.

6.3 Geometric Deep Learning

Bronstein et al. [1, 2] established the "5G" framework for geometric deep learning: graphs, grids, groups, manifolds, and gauges. This builds upon earlier foundational work on Group

Equivariant Convolutional Networks [5], which first demonstrated the power of explicitly encoding symmetries in neural architectures. Our work extends this lineage by proposing that cognitive domains should be modeled as manifolds with invariant structures governing valid transitions.

We differ from prior work in our focus on conservation laws as the fundamental organizing principle, in our claim that this approach enables native multimodality through geometric unification, and in our distinction between discrete integration (G-SSM) and continuous flow dynamics (ISN) as two realizations of the same paradigm.

7 Empirical Validation

While the preceding sections established the theoretical foundations of the GFN paradigm, we present empirical evidence from its two primary realizations: the **Geodesic State Space Model (G-SSM)** and the **Inertial State Network (ISN)**. These experiments validate that the theoretical framework translates into robust inductive biases and stable execution.

7.1 Realization A: Geodesic State Space Model (G-SSM) Dynamics

The **G-SSM** demonstrates the framework’s capability to capture task-specific algorithmic structures through geometric symmetry, achieving significant out-of-distribution extrapolation.

7.1.1 Cumulative XOR Parity

We evaluate the framework’s ability to learn abstract logical symmetries. The task computes the cumulative parity (XOR) of a bitstream. While a statistical model (Transformer) struggles with sequences beyond its training context due to positional interpolation limits, the **G-SSM** learns the underlying toroidal symmetry that encodes parity conservation.

Table 2: XOR Logic Convergence and Extrapolation. Model trained on $L = 20$ tokens.

System	Training Steps	Test Length (L)	Accuracy
G-SSM (Rel. A)	<200	1,000,000	100%

Interpretation: The **G-SSM** achieves perfect extrapolation to 1,000,000 tokens after training on only 20, reaching 100% accuracy in under 200 training steps. This confirms that the model has not learned patterns but the **physics of the parity operation** on a toroidal manifold. The particle flows along geodesics that naturally preserve parity, enabling generalization that is not merely statistical interpolation but geometric necessity.

7.1.2 Information Retrieval: Multi-Needle-in-a-Haystack

Task: Detect and remember multiple target signals (“needles”) dispersed throughout a long non-relevant context (“haystack”), then output only after all targets have been observed.

Setup: Train with $K = 2$ needles in $L = 64$ token context. Test with context lengths up to 32,000 tokens: a 500-fold increase over training context.

Table 3: Multi-Needle-in-a-Haystack results. Model trained on $L = 64$ and tested up to $L = 32,000$.

Context Length	Accuracy	False Positive Rate	Needle Separation
1,000	100%	0.0%	0-500 tokens
4,000	100%	0.0%	0-2,000 tokens
16,000	100%	0.0%	0-8,000 tokens
32,000	100%	0.0%	0-16,000 tokens

Interpretation: The zero false positive rate is particularly significant. A statistical model might learn to output “1” after observing certain local patterns that correlate with needle presence, but would fail when those patterns are absent or when needles are separated by large distances. The geometric model maintains state because needle events leave persistent imprints on the phase space through momentum conservation, not because it has learned to recognize statistical proxies. This demonstrates **inductive persistence**: the ability to maintain information about events arbitrarily far in the past without degradation.

7.1.3 Modality-Agnostic Processing: Drone Detection

To validate the modality-agnostic claims of Section 1.2, we test the **G-SSM** on a fundamentally different modality: real-time drone detection from continuous video streams.

Task: Detect and track drone objects in video sequences.

Table 4: G-SSM drone detection results demonstrating modality-agnostic processing (Preliminary Benchmarks).

Mode	Parameters	VRAM	Throughput
Training	160,000	80 MB	—
Inference	160,000	<80 MB	80 FPS

Interpretation: This result validates that the **G-SSM** architecture processes diverse modalities through the same underlying mathematical engine. To bridge the visual signal to the manifold’s force space $\mathbb{R}^D$, we utilize a **Sensory Adapter** (a Patch-based Image Projector). While this adapter is modality-specific (analogous to the retina or auditory nerve), the **Geodesic Engine** (the “brain”) remains identical across all tasks. The same inertial dynamics govern the state evolution regardless of whether the force impulses originate from text or pixels.

7.2 Realization B: Inertial State Network (ISN)

The **ISN** instantiates the GFN paradigm for high-fidelity world state persistence through continuous geodesic flow, demonstrating that physics-based simulation scales to complex tasks like language generative modeling.

7.2.1 Language Modeling: Shakespearean Simulation

We apply the **ISN** to character-level language modeling using the TinyShakespeare dataset. Unlike standard LLMs that map tokens to tokens through statistical association, the **ISN** treats sentences as trajectories of state flowing through an internal world geometry, where reasoning emerges from geodesic dynamics rather than pattern completion.

Table 5: ISN Language Performance (Epoch 29) on TinyShakespeare.

Metric	ISN (Rel. B)	nanoGPT (Under-scale)
Parameters	363,329	369,504
Perplexity (PPL) ↓	2.48	4.68
World Coherence ↑	0.980	N/A
Training Loss	0.966	1.546
Inference Latency	$O(N)$	$O(N)$
VRAM Usage ($L = 32$)	13.68 MB	13.69 MB

Comparison highlights the architectural capability in the ultra-low parameter regime. **nanoGPT** is benchmarked here under the same parameter constraints (340k) to isolate the efficiency gap; while nanoGPT fails to attain grammatical coherence at this scale, the ISN maintains stable Shakespearean flow. [†]Measured on our modular implementation; includes telemetry buffers. [‡]State memory requirements. Total VRAM for nanoGPT at $L = 8192$ with Flash Attention is 230.70 MB, compared to 228.54 MB for ISN- $O(1)$.

Interpretation: Achieving a perplexity of 2.48 demonstrates that geometric approaches can significantly outperform statistical baselines in the ultra-low parameter regime. We utilize **nanoGPT** (369k parameters) as a representative Transformer baseline for this comparison; at this extreme scale, the Transformer fails to attain grammatical coherence, while the ISN maintains stable Shakespearean flow. The primary advantage is the **World Coherence** metric (0.980), which measures the stability of the internal world state throughout generation.

The VRAM efficiency is particularly notable in persistent generative tasks. While Transformers require a **KV-Cache** that scales linearly with context length ($O(L \cdot d \cdot n_{layers})$), the ISN only requires the **persistent state vector** ($O(d)$). At $L = 32,768$ and a 10M parameter scale, nanoGPT requires **240.0 MB** for its KV-cache, whereas the ISN requires only **3.0 KB** for its internal state—a **80,000×** reduction in *state persistence memory*. It is important to distinguish this from total peak VRAM in batch mode, where both architectures are bounded by input/gradient buffers.

7.3 Memory and Stability

Beyond task performance, we validate the fundamental architectural properties: constant memory scaling, native multimodality, and structurally-grounded coherence.

7.3.1 Memory Scaling Analysis

To validate the $O(1)$ memory claim, we measure VRAM usage as a function of sequence length:

Interpretation: All historical information is encoded in the phase space configuration (G-SSM) or geodesic trajectory (ISN), not in stored activations. Consistent with our

Table 6: Memory scaling (363k parameters) representing the $O(1)$ trend.

Length (L)	State Memory	Total VRAM (ISN)	nanoGPT (369k)
32	2.00 KB	13.68 MB	13.69 MB
1024	2.00 KB	13.68 MB	13.70 MB
2000	2.00 KB	13.68 MB	13.72 MB

At this scale, the persistent world-state requires strictly 2.00 KB. The constant Total VRAM for ISN demonstrates that the internal dynamics do not scale with context during inference, whereas the Transformer baseline exhibits an $O(N)$ growth even in its 'light' configuration.

Table 7: Agnostic Architectural Scalability (10M Parameters Scale)

Context (L)	State Memory	Total VRAM (ISN)	nanoGPT (10M)
32	3 KB	137.42 MB	141.65 MB
1024	3 KB	137.42 MB	152.66 MB
8192	3 KB	137.42 MB	230.70 MB
Inference Speed (GTX 1650)	700 TPS	$O(1)$ Step Cost	$O(N^2)$ Re-processing

While the *Total VRAM* (including input buffers) scales linearly in current PyTorch implementations, the **State Memory** remains strictly constant at 3 KB. Crucially, the inference speed is context-independent (**700 TPS**) due to the $O(1)$ per-token complexity, whereas attention-based models suffer significant latency degradation ($O(N^2)$ total, $O(N)$ per step).

theoretical model and empirical audit, VRAM consumption remains strictly bounded even at context lengths that would typically require massive distributed memory systems. The observed constants represent an order-of-magnitude reduction in infrastructure overhead compared to attention-based models.

7.3.2 Hallucination and State Stability

We empirically validate the theoretical claim that geometric models eliminate hallucination through hard constraints. We evaluate the deterministic stability of the **G-SSM** realization on the toroidal manifold, measuring hallucination as the spontaneous activation of the positive class (class=1) in the absence of any trigger signal. Concretely, we feed the model sequences of all-zero tokens representing pure haystack with no needles, and measure the toroidal distance of the phase state $\mathbf{x}$ from the target angles $\pm\pi/2$ that correspond to class 0 and class 1 respectively. The Hallucination Score is defined as:

$$HS = \max_{b,t} \frac{e^{-d_1(b,t)}}{e^{-d_0(b,t)} + e^{-d_1(b,t)}} \quad (13)$$

where $d_k(b,t)$ is the mean toroidal distance of $\mathbf{x}$ from the class- k target angle at batch b , timestep t . A score near 0.5 indicates geometric neutrality; a score near 1.0 indicates the model spontaneously drifts toward the positive class.

Interpretation: The near-zero hallucination scores (0.0456 maximum, 0.043 mean) indicate that the geometric framework produces deterministic, reproducible state trajectories. This is direct: the flow is governed by the **Topological Boundary** of the manifold: in a

Table 8: Hallucination and stability metrics for geometric framework.

Metric	Value
Hallucination Score (max)	0.0456
Hallucination Score (mean)	0.043
State Drift per Step	0.001858
Temporal Variance	0.0083
Inference Time (100 tokens)	0.604 s
Parameters (Core / Task)*	3,164 / 7,248

*Core parameters define the invariant manifold geometry; Task parameters handle input/output projections.

categorical domain (e.g., bitwise XOR), the manifold is a Torus ($\mathbb{T}^d$), and the rules of the domain are encoded as discrete symmetries of this topology. In more complex semantic domains (e.g., Natural Language), the framework assumes a symmetry of **Semantic Translation Invariance**, where meaning remains locally invariant under state transitions, giving rise to the conservation of semantic momentum as predicted by Noether’s Theorem. In logical domains like XOR, violations are geometrically inconsistent with the manifold topology, effectively eliminating categorical hallucinations.

However, we distinguish this from **Metric Precision** in high-dimensional semantic spaces (e.g., language). While the ISN provides structural grounding, the “hardness” of its resistance is a function of the learned curvature resolution. At the current 363k scale, we observe **Geodesic Context Mixing**: a phenomenon where the state trajectory enters overlapping thematic regions due to insufficient manifold separation. Unlike statistical hallucinations, context mixing is a deterministically traceable path error caused by limited metric precision, not a stochastic failure.

7.4 The Inductive Bias Argument

The collective evidence from these experiments supports a fundamental claim about the nature of learning in geometric models: **they learn problems, not patterns**.

When a model trained on $L = 20$ extrapolates to $L = 1,000,000$, it has not merely demonstrated generalization; it has captured the underlying algorithmic structure. When a model tracks needles separated by 16,000 tokens without degradation, it does not remember the needles through statistical association; it encodes them in dynamical invariants that persist indefinitely through geodesic flow.

This is the inductive bias we described in Section 2.3. Conservation laws are not soft regularization terms; they are hard constraints that force the model to represent the causal structure of the domain. The particle cannot flow into invalid states because the geometry of the manifold does not include them.

This stands in contrast to attention-based models, which learn statistical correlations that may hold in the training distribution but fail outside it. The geometric approach learns the physics of the domain: the rules that govern valid state transitions. The particle flows along geodesics that exist independently of training data, enabling robust generalization to distributions not seen during training.

8 Experimental Setup and Hyperparameters

We provide detailed hyperparameters and architectural specifications for reproducibility. Source code is available at <https://github.com/DepthMuun/gfn>, and checkpoints/logs can be found at huggingface.co/DepthMuun.

8.0.1 Configuration and Evaluation Details

All experiments were conducted on consumer-grade hardware to demonstrate the architectural efficiency of our framework. Testing and convergence benchmarks were performed on a system equipped with an NVIDIA GeForce GTX 1650 (4GB VRAM), an AMD Ryzen 5 3500 6-Core processor, and 16GB of system RAM.

This hardware choice serves as a practical baseline for demonstrating that physics-informed state evolution does not require industrial-scale clusters to achieve long-context stability. For sequence lengths up to 32,000 tokens, our realizations maintain a peak VRAM footprint of approximately 203 MB, representing an order-of-magnitude reduction in infrastructure overhead compared to optimized Transformer implementations using FlashAttention and adaptive KV-caching.

Detailed hyperparameter settings and hardware-specific configurations for all experimental runs are provided in Appendix C (Section C).

9 Discussion

9.1 Implications for AI Development

The empirical validation of these theoretical propositions dictates a fundamental realignment in the trajectory of AI research:

- **From Scaling to Structure:** Current research emphasizes scaling compute and data. Geometric intelligence underscores the importance of structuring the computational substrate to encode domain physical invariants.
- **From Hardware to Algorithms:** Current hardware trends emphasize larger GPUs and more memory. Geometric intelligence suggests that algorithmic improvements, specifically better manifold construction and flow dynamics, may yield larger returns than hardware scaling.
- **From Statistical to Causal:** The shift from probability to conservation fundamentally changes what models can reason about. Geometric models reason about causation because they model causal structure directly through invariant-preserving dynamics; statistical models reason about correlation because they lack the geometric grounding necessary for causal inference.

This perspective relates to the discussion on **Scaling Laws** [18], which characterizes the empirical relationship between compute, data, and performance in large-scale models. While scaling has yielded unprecedented progress, geometric intelligence suggests that architectural priors, specifically the encoding of domain physics through geodesic flow and conservation laws, may offer a complementary path toward more efficient and robust intelligence.

9.2 Limitations and Challenges

We acknowledge several limitations of our proposed framework:

- **Training and Inference Speed:** The sequential nature of geometric dynamics precludes the parallel token processing that enables Transformers to train efficiently on GPUs. Forward and backward passes must proceed through the flow, resulting in training times that scale linearly with sequence length rather than constant-time parallelization.
This is the dominant computational bottleneck in most practical applications. While inference memory is $O(1)$ and dramatically lower than attention-based models, inference time remains sequential. For many tasks, this tradeoff is acceptable (inference memory often dominates cost at scale), but it is a fundamental limitation that must be acknowledged.
- **Training Stability:** Geometric models require learning the manifold structure, which is computationally challenging. The optimization landscape for manifold learning is less understood than for attention-based models, requiring careful initialization and regularization.
- **Expressiveness:** Hard constraints enforce validity but may limit expressiveness in domains where valid outputs are not easily characterized by conservation laws or geodesic flows.
- **Scalability and Precision:** While theoretical complexity is favorable, practical realizations require high metric precision to avoid context mixing.

We view these challenges as research opportunities. Specifically, we distinguish between **Topological Boundaries** (XOR) and **Metric Plasticity** (Language). If a geometric model produces an error, it is typically a consequence of a mislearned or under-resolved manifold geometry—a deterministically reproducible path that is auditable through curvature visualization. In the current ISN realization, **Geodesic Context Mixing** represents the primary failure mode, where the flow traverses adjacent thematic regions. This contrast favors GFN for mission-critical intelligence where auditability is more valuable than erratic statistical accuracy.

9.3 The Broader Perspective

We end with a broader observation. The current AI landscape is dominated by scaling laws: empirical relationships between compute, data, and performance. These scaling laws suggest that current approaches will continue improving for the foreseeable future. However, they also suggest that continued improvement will require continued scaling of resources.

We argue that the field has reached the point where the cost of scaling has become prohibitive, not in absolute terms but in terms of marginal return on investment. The next paradigm shift will not come from better scaling but from a fundamental change in approach.

While the attention mechanism maximized the utility of statistical pattern matching, the geometric framework unlocks the next era: intelligent systems grounded in determinism and physical invariants. This is not merely a logical progression but the necessary structural evolution to achieve long-horizon, reliable intelligence. The particle flowing

through manifold curvature provides a more stable foundation than statistical correlation because geometry exists independently of training distribution.

10 Conclusion

We have proposed an alternative: the **Geometric Flow Networks (GFN)** paradigm, which replaces statistical correlation with conservation laws, quadratic attention with geodesic flow, and probabilistic sampling with deterministic dynamics. This approach offers demonstrated advantages in computational complexity (achieving iterative constant-memory inference regardless of context length), multimodal integration, and out-of-distribution extrapolation.

We have presented both theoretical foundations and empirical validation demonstrating that the geometric framework achieves:

- **Algorithmic Extrapolation:** **G-SSM** trained on $L = 20$ achieves 100% accuracy on XOR at $L = 1,000,000$ ($50,000\times$ extrapolation) with convergence in under 200 steps.
- **Information Retrieval:** Perfect accuracy (100%) on Multi-Needle-in-a-Haystack with 32,000 token separation, demonstrating inductive persistence through momentum conservation.
- **Language Simulation:** **ISN** achieves a perplexity of **2.48** on TinyShakespeare with only **363,329 parameters**, demonstrating superior parameter efficiency compared to Transformer baselines at the same scale.
- **Inference Throughput:** Constant-memory state updates enable a stable throughput of **700 TPS** at $L = 2000$ (on GTX-class hardware), bypassing the $O(N^2)$ re-processing requirement of stateless attention.
- **Memory Complexity:** **Demonstrated constant** inference memory (**13.68 MB** at $L = 2000$) for the full system, while the persistent world-state remains strictly constant at **2.00 KB**.
- **State Stability:** Hallucination score of 0.0456 (near-zero) and World Coherence of **0.980**, demonstrating deterministic, physically-grounded generation through invariant preservation.
- **Modality-Agnostic Dynamics:** A unified engine processes diverse signals, including text and visual streams, through consistent geodesic dynamics.

While engineering optimizations remain to scale throughput, the empirical evidence suggests that intelligence does not inherently require quadratic memory scaling. The geometric framework provides a parameter-efficient alternative to the statistical paradigm, offering a structural path toward long-horizon sequential modeling.

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A Appendix A: The G-SSM Realization: Theoretical Mechanics

In this section, we provide further architectural details regarding the **Geodesic State Space Model (G-SSM)** implementation. The central innovation of this realization is the implementation of the GFN paradigm as a second-order manifold state space model. While first-order SSMs (e.g., S4, Mamba) evolve a hidden state h_t through linear or selective transitions, **G-SSM** evolves a physical state in phase-space, defined by position $\mathbf{x}$ and velocity $\mathbf{v}$. This enables the model to capture not just the state but the momentum of semantic transitions, where inertia carries forward information from arbitrarily distant past.²

A.1 Architectural Blueprint

The **G-SSM** layer processes tokens by treating them as force inputs. The update logic is governed by predicted Christoffel symbols Γ_{jk}^i , which define the local curvature of the semantic space. State flows along geodesics determined by this curvature, with external forces bending the trajectory without replacing the configuration. Detailed architecture is provided in Figure 1.

B Appendix B: The ISN Realization: Continuous Flow Mechanics

The **Inertial State Network (ISN)** serves as a continuous dynamic extension of the GFN paradigm, focused on maintaining a persistent world-state through continuous geodesic flow rather than discrete state transitions. Unlike standard autoregressive models that map tokens to tokens, the **ISN** treats information as being lived by state flowing through manifold curvature. The past shapes the future not through accumulation but through the way history has molded the geodesic trajectory.³

The ISN fundamentally differs from G-SSM in its treatment of time. While G-SSM uses discrete symplectic integration with fixed timesteps, ISN eliminates discrete timesteps entirely, treating state evolution as a continuous differential flow governed by inertial momentum and geometric coupling. This enables three capabilities unavailable in discrete models: (1) querying state at arbitrary temporal resolutions (e.g., $t = 1.5$), (2) asynchronous input processing without global clock synchronization, and (3) inherent persistence without explicit memory management.

B.1 Operational Pipeline

The ISN’s lifecycle follows a streamlined three-module pipeline:

²The “state as particle” description is metaphorical; the essential property is second-order dynamics with persistent phase-space variables.

³“State flowing” is literal in ISN: the continuous differential equation $\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, u(t))$ replaces the discrete recursion $h_{t+1} = f(h_t, x_t)$.

1. **Scanning:** The **GFNScanner** encodes input tokens through an embedding layer followed by flow dynamics $\tanh(W \cdot s)$, producing impulse sequences that preserve semantic momentum.
2. **Geodesic Evolution:** The **GFNWorld** evolves the persistent state $\mathbf{W}$ through drift (inertia) and diffusion (external impulses). The update rule $\mathbf{W}_{t+1} = \mathbf{W}_t + \text{drift}(\mathbf{W}_t) + \text{diffusion}(\mathbf{f}_{ext})$ ensures $O(1)$ memory complexity through state persistence.
3. **Materialization:** The **ThresholdEmitter** projects world states to token logits through learned thresholding, completing the pipeline from geometric state to discrete output.

B.2 Internal Architecture Diagram

The detailed internal structure of the **ISN**, showing the three-module architecture, is illustrated in Figure 2.

B.3 World Coherence Metric

World Coherence ($\mathcal{C}$) measures the stability of the internal world state $\mathbf{W}$ throughout a generation trajectory. The state $\mathbf{W}_t \in \mathbb{R}^D$ represents the current configuration of the persistent world (not a collection of entities). The coherence is defined as the inverse of the normalized state drift (Equation 14):

$$\mathcal{C} = 1 - \frac{1}{T} \sum_{t=1}^T \frac{\|\mathbf{W}_t - \mathbf{W}_{t-1}\|_2}{\|\mathbf{W}_t\|_2 + \epsilon} \quad (14)$$

where T is the sequence length and ϵ is a small stability constant. A value near 1.0 indicates that the model maintains a persistent and globally consistent internal state through continuous geodesic flow. This metric directly measures compliance with the **Persistent Internal World** and **Structural Invariance** pillars of the GFN paradigm, demonstrating that the internal world exists as geometry rather than as a memory buffer storing historical activations.

C Appendix C: Hyperparameters and Hardware Configuration

In this section, we consolidate the technical configurations used in all experimental runs to ensure reproducibility across a variety of computational environments.

C.1 G-SSM (Realization A) Configurations

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Table 9: XOR Logic Convergence and Extrapolation settings (Table 2).

Parameter	Value
<i>Model Architecture</i>	
Vocab Size	2
Dimension (d)	8
Depth	1
Heads	2
<i>Physics Configuration</i>	
Topology	Torus
Embedding Type	Functional (Linear)
Coordinate Dimension	16
Impulse Scale	80.0
Integrator	Yoshida
<i>Stability</i>	
Base dt	0.4
Velocity Saturation	10.0
Numerical Hysteresis	0.05
Trace Normalization	Enabled

3pt

Table 10: Multi-Needle-in-a-Haystack experimental setup (Table 3).

Parameter	Value
<i>Model Architecture</i>	
Dimension (d)	16
Depth	2
Heads	2
Integrator	Leapfrog
<i>Physics Configuration</i>	
Topology	Torus (Low-Rank Riemannian)
Coordinate Dimension	16
Impulse Scale	80.0
Reactive Curvature Plasticity	0.2
<i>Training</i>	
Train Seq Length	64
Train Batch Size	64
Train Steps	800
Learning Rate	1×10^{-3}

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Table 11: Visual Stream Detection hyperparameters (Table 4).

Parameter	Value
<i>Model Architecture</i>	
Image Size	64×64
Patch Size	32×32
Dimension (d)	32
Rank	32
<i>Physics Configuration</i>	
Topology	Torus (Low-Rank)
Impulse Scale	80.0
Numerical Hysteresis	0.2
Base dt	0.1

C.2 ISN (Realization B) Configuration

2pt

Table 12: ISN Shakespeare Hyperparameters (Consolidated).

Model Config	Value	Optimization	Value
Total Parameters	363,329	Optimizer	AdamW
d_{model}	256	Learning Rate	5×10^{-4}
N Layers	8	Persistence	Deep Stateful Pipeline
$d_{\text{embedding}}$	256	Batch Size	512
$d_{\text{properties}}$	128	Context Length (L)	128
Integrator	GFNWorld	Gradient Clip	1.0
Loss Weights	Value	Curriculum Learning	Epochs
λ_{outcome}	1.0	Phase 1 ($\lambda_{\text{coh}} = 0.5$)	0–10
$\lambda_{\text{coherence}}$	1.0	Phase 2 ($\lambda_{\text{coh}} = 1.0$)	10–50
$\lambda_{\text{efficiency}}$	0.2	Total Epochs	50